MANAGEMENT OF EARLY NEONATAL JAUNDICE (NNJ) IN THE CHILDREN'S EMERGENCY

AIM

This guide serves to supplement the "Guidelines for Management of Neonatal Jaundice" by the Department of Neonatology, National University Hospital. It focuses on initial assessment of the neonate referred to the Children's Emergency for evaluation of early neonatal jaundice. This guideline is based off the Clinical Practice Guidelines titled "Guidelines on Evaluation and Management of Neonatal Jaundice" by the College of Paediatrics and Child Health, Singapore (CPCHS, October 2018).

For patients with G6PD deficiency with early NNJ, please refer to "Management of Jaundice in G6PD deficiency" under Neonatology departmental guidelines (available on intranet).

BACKGROUND AND DEFINITONS

Jaundice is caused by accumulation of bilirubin in the skin and mucous membranes as a result of hyperbilirubinemia.

Pathophysiology

1. Increased bilirubin production: Newborns have more RBCs with shorter lifespans. Increased turnover of RBCs produces more bilirubin
2. Decreased hepatic clearance: Immature liver with decreased activity of uridine diphosphogluconurate glucorunosyltransferase (UGT1A1)
3. Increased enterohepatic recirculation of bilirubin results in increased bilirubin load

Definitions:

Hyperbilirubinemia	= SB > 205umol/L
Moderate hyperbilirubinemia	= SB > threshold for phototherapy
Severe/extreme hyperbilirubinemia	= SB > threshold for exchange transfusion
Physiologic rate of rise	= 60 – 80 umol/L per 24H
Pathological rate of rise	= > 103 umol/L per 24H

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Dr Ian Tan

Causes of NNJ

1. Physiological jaundice
2. Breastfeeding failure jaundice
   The failure of successfully initiating and establishing breastfeeding plays a major role. Though the exact mechanism is unknown, it is likely to be multi-factorial involving decreased calorie intake, inhibition of hepatic bilirubin excretion, increased intestinal bilirubin reabsorption.
3. Increased hemolysis
   i. Polycythemia
   ii. ABO incompatibility with hemolysis (DCT positive)
   iii. G6PD deficiency
   iv. Isoimmune hemolysis (eg. hereditary spherocytosis etc.)
4. Sepsis / Intra-uterine (TORCH) infections
5. Inborn errors of metabolism

Evaluation of NNJ

Prior to the consult it would be useful to know:

<table>
<tr><td colspan="3">Gestational age (GA)</td></tr>
<tr><td>Preterm</td><td>Below GA 35
(GA 34^{+6} and less)</td><td>Weight based threshold
(TCB not recommended)</td></tr>
<tr><td>Late preterm</td><td>GA 35^{+0} – 36^{+6}</td><td>High risk</td></tr>
<tr><td>Early term</td><td>GA 37^{+0} – 38^{+6}</td><td rowspan="2">Refer to standard table of threshold values</td></tr>
<tr><td>Term</td><td>GA 39^{+0} – 40^{+6}</td></tr>
<tr><td>Late term</td><td>GA 41^{+0} – 41^{+6}</td><td rowspan="2">At risk for birth complications – birth trauma (cephalohematoma)
Chronic hypoxia – polycythemia</td></tr>
<tr><td>Post term</td><td>GA 42 and above</td></tr>
<tr><td colspan="2">Mode of delivery</td><td>Assisted delivery / instrument delivery increased risk of birth trauma (cephalohematoma / bruising)</td></tr>
<tr><td colspan="2">% weight loss from BW
% change from last weight

BW = birthweight
DW = discharge weight
LW = last weight
CW = current weight</td><td>Un-supplemented breastfed infants experience maximum weight loss by D3 – 4 (ave. 4 – 10% loss) and regains BW by D10 – D14.
If weight loss ≥ 10% – detailed feeding history must be taken

❖ Look at rate of weight loss by trending BW, DW/LW and CW
❖ If weight loss between LW/DW to CW is significant – it is a surrogate measure of quality of feeding at home</td></tr>
<tr><td colspan="2">Cord blood screening results (G6PD and cord *TSH)

*hypothyroidism associated with prolonged jaundice</td><td>G6PD deficiency is associated with rapid rise of bilirubin:
a. increased lysis of RBC
b. associated with co-inheritance of variant UGT1A1 (decreased hepatic clearance)

<u>Threshold values</u> for phototherapy are <u>LOWER</u> for those with G6PD deficiency. (Refer to neonates guideline on intranet for management of jaundice in G6PD deficiency)</td></tr>
<tr><td colspan="2">Blood group of mother / baby</td><td><u>ABO setup</u>
<table>
<tr><td>Mother’s blood group O</td><td><u>AND</u></td><td>Baby’s blood group
- unknown, <u>OR</u>
- A, B, or AB</td></tr>
</table>

<u>ABO incompatibility</u>
<table>
<tr><td>ABO setup</td><td><u>AND</u></td><td>DCT positive, <u>OR</u>
Reticulocyte count > 8%</td></tr>
</table>

Baby’s risk classification is <u>revised</u> when:
o Baby’s blood group is known (Blood group O) <u>OR</u>
o Baby is more than 48H of life (in the absence of a Direct Coomb’s Test, any hemolysis is presumed minimal/negligible if child does not show clinical signs of hemolysis – pallor, severe hyperbilirubinemia with symptoms of anemia)</td></tr>
</table>

History

Duration of jaundice	≤ 14 days	Early neonatal jaundice
	> 14 days	Prolonged neonatal jaundice
Maternal history	Set up for sepsis (peri-partum fever, GBS status, PROM > 18H) Adequate IAP (= at least 1 dose of ampicillin administered 4H prior to delivery) Features of chorioamnionitis (foul smelling liquor)	
	Maternal ingestion of Traditional Chinese Medication (TCM) or drugs/medication	
Family history	Previous child receiving phototherapy within first 24H of life or requiring exchange transfusion	
Feeding history		
Type of feeds	Breastfeeding	Breastfed babies are more likely than bottle-fed babies to develop physiological jaundice within the first week of life. Prolonged jaundice is also more commonly seen in breastfed babies.
	Formula milk	
Adequacy of feeds	Breastfeeding	☐ Engorgement of breasts prior to feed, with ☐ Let down of milk with emptying of breasts with relief of pressure ☐ Sounds of baby sucking/swallowing milk ☐ Degree of satiety post-feeding (2 – 3H of sleep before waking) ☐ Duration of breastfeeding approximately 30 minutes (max.) – if duration is longer, suspect non-nutritive suckling.
	Formula milk	☐ Degree of satiety post-feed ☐ Ave. feed volume (BW used for calculation until child crosses it) D1 = 60 ml/kg per day D2 = 80 ml/kg per day D3 = 100 ml/kg per day D4 = 120 ml/kg per day ≥ D5 = 120 – 150 ml/kg per day
	Diaper change (WD = wet diapers)	☐ D1 – D3: At least 1WD for each DOL (eg. 1WD on D1; 3WD on D3) ☐ ≥ D4: Average of 4 to 6 WD per day
	Color of stools	☐ Average 3 to 4 stools per day (by D4) ☐ Stool color transitions from meconium to mustard yellow mushy stools (by D3 – D4) ❖ Persistent meconium – suggests inadequate feeding with increased enterohepatic recirculation of bilirubin ❖ Pale stools – suggest obstructive cause of jaundice

Physical Examination

General	Activity / reactivity of infant to touch	
Hydration	Fontanelles, capillary refill (central)	
Visual assessment of jaundice **modified Kramer scale*	Level by anatomical level (jaundice progresses cephalo-caudally)	
	Face	100 umol/L
	Chest – umbilicus	150 umol/L
	Umbilicus – knees	200 – 250 umol/L
	Arms & calves	250 – 300 umol/L
	Hands & feet	>300 umol/L
Signs that suggest pathological causes	Cephalohematoma Extensive bruising	Extravascular reservoir of RBC which release bilirubin as they breakdown
	Hemolysis	Pallor, splenomegaly, dark urine in diapers
	Sepsis	Skin mottling, poor capillary refill, bulging fontanelles
Acute bilirubin encephalopathy	Early	Lethargy, hypotonia, poor suck
	Intermediate	Moderate stupor, irritability, hypertonia (retrocollis & opisthotonos) , fever, high-pitched cry
	Advanced	Pronounced retrocollis-opisthotonos, shrill cry, apnea, deep stupor, seizures
Well-child examination	CNS	Moro's reflex, presence of jitteriness
	CVS	Presence of murmurs
	GI	Umbilical stump health Umbilical / Inguinal hernias Patency of anus
	Genito-urinary	Ambiguous genitalia Boys – state of descent of testes
	MSK	Spine for dysraphism

Disposition

Admission

(location dependent on SB levels – Refer to Figure 1.)

Neonatal department

i. Nursery (well infants requiring phototherapy)
   - Single Blue Light (SBL) – SB level crosses phototherapy threshold
   - Double Blue Light (DBL)
     - SB level within 50umol/L of exchange transfusion threshold
     - Rate of rise of SB > 5umol/L per hour

ii. Neonatal Intensive Care Unit (NICU)/Special Care Nursery (SCN)
   - Intensive phototherapy – SB level within 25umol/L of the exchange transfusion level
   - SB level crosses thresholds for exchange transfusion

Pediatric department

iii. Pediatric General Ward (unwell infants)
   - Fever and/or infective symptoms
   - Abnormal vital signs
   - Dehydration requiring intravenous hydration

Discharge

i. With follow up
   - To predict what SB level will be in 24H
     - Calculate rate of rise using last 2 SB values and add to current value
     - If value is within 25umol/L of threshold – TCU 1 day
     - If value is less than 25umol/L of threshold – TCU 2 day

$$\underset{(\textit{calculated value})}{\text{SB rise per 24H}} = \frac{(\text{SB}_{\text{current}} - \text{SB}_{\text{last}})}{(\text{No. of hours between SB}_{\text{current}} \text{ and SB}_{\text{last}})} \times 24\text{H}$$

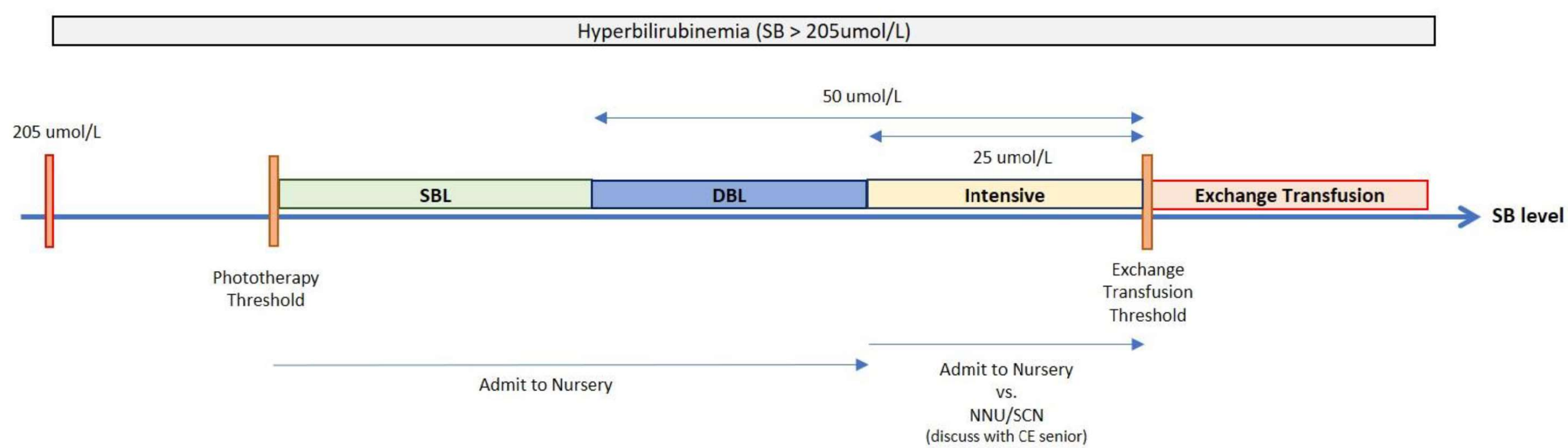
   - Refer to template below for memo to provide parent for TCU

ii. With no follow up

Patients can be discharged with no follow up for recheck of SB levels if the following conditions are fulfilled.

1. Last 2 SB values downward trending with no intervention
2. Feeding established with good weight gain

Figure 1.



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Frequently Asked Questions

1. **What is phototherapy? Why can't I put my child under the morning sun?**

Phototherapy is the safest and most common intervention used to treat high levels of jaundice in term and late preterm babies. It involves exposing the baby to light of a specific wave-length (hence the blue color) which reduces jaundice levels in the blood. It does so by converting the toxic bilirubin pigments into a less-toxic and more water soluble compound so that it can be passed out through the bile and excreted through the urine readily. It is the liver's job to process the insoluble toxic bilirubin pigments to non-toxic soluble ones. Like how your baby is slowly adjusting to being in this new environment, so is your baby's liver function which will eventually pick up in days to come as it matures with baby.

Sunlight is able to lower baby's jaundice levels but for it to effectively do so, this would mean exposing a large proportion of your baby's skin to the sunlight for a sufficient length of time. This comes with the risk of raised body temperatures, exposure to harmful UV rays and sun burns and dehydration.

2. **What are the side effects of phototherapy?**

Phototherapy is extremely safe and its complications are rare and usually temporary. Some of the most common complications are:

i. Dehydration and overheating
ii. "Bronze baby syndrome" – dark grayish brown discoloration of the skin and urine. It is usually not harmful and gradually resolved without treatment after several weeks.
iii. Skin rash
iv. Loose bowel movements

Jaundice Management Guidelines (≥ GA 35 and BW >2000g)

Term babies and well, late-preterm babies

THRESHOLD VALUES

PHOTOTHERAPY

Age	Normal Risk SB (TcB) umol/L	High Risk SB (TcB) umol/L
0 – 12H 13 – 24H		100 (90) 150 (135)
24 – 36H 37 – 48H	200 (180) 225 (200)	175 (160) 200 (180)
49 – 72H 73 – 96H	250 (225) 275 (250)	225 (200) 250 (225)
97 – 120H	275 (250)	250 (225)
121H – 7d	300 (270)	275 (250)
8d – 14d	325 (≥300)	300 (270)

TcB – Transcutaneous Bilirubin threshold for performing SB
SB – Serum Bilirubin threshold for initiation of phototherapy

EXCHANGE TRANSFUSION

Age	Normal Risk SB umol/L	High Risk SB umol/L
0 – 12H 13 – 24H		200 250
24 – 36H 37 – 48H	300 350	275 300
49 – 72H 73 – 96H	350 375	325 350
97 – 120H	400	350
121H – 7d	400	375
8d – 14d	425	375

Normal Risk		No "High Risk" risk factors ▪ Term (GA ≥ 37) ▪ AGA (BW > 2500g)
High Risk	Risk for severe hyperbilirubinemia	▪ Family hx of severe NNJ (eg. siblings required exchange tx) ▪ Late preterm (GA 35^{+0} – 36^{+6}) ▪ SGA (BW < 2500g) with IUGR ▪ TBF with weight ≥ 10% below BW ▪ Visible jaundice within 24H of life ▪ Isoimmune hemolytic anemia (eg. Hereditary spherocytosis) ▪ (Mother O and Baby A,B or AB) **AND** (positive DCT or reticulocyte count > 8%) ▪ Rhesus incompatibility ▪ Rate of SB rise > 103umol/L per day
	Risk of bilirubin crossing BBB	▪ Perinatal asphyxia (Apgar ≤ 5 @ 5 min of life) ▪ Sepsis ▪ Acidosis ▪ Serum albumin < 30g/L

**Adapted from Guidelines on Evaluation and Management of Neonatal Jaundice (October 2018) by College of Paediatrics and Child Health, Singapore (CPCHS)*

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